

HPE Alletra Storage MP B10000 QuickSpecs

HPE Alletra Storage MP B10000 provides mission-critical storage at midrange economics with the industry's first disaggregated, scale-out block and file storage.

Built on HPE Alletra Storage MP hardware and managed via the HPE GreenLake cloud, this unique unified block and file storage offering brings an intuitive cloud experience, efficient scale, and extreme resiliency and performance to mission-critical apps from large databases to modern cloud-native apps to consolidated mixed workloads at a midrange price point.

Overview

Get an intuitive, AI-driven cloud experience on-premises that simplifies management and shifts operations from infrastructure-centric to app-centric. Scale performance and capacity independently and without disruption with disaggregated storage. Meet every SLA with mission-critical storage that delivers an unrivaled 100% data availability guarantee.



HPE Alletra Storage MP B10000
(2-Node all NVMe Storage Base)

What's New

- The new HPE Alletra Storage MP B10140 (4N) is available in a switchless, 4-node configuration that delivers lower total cost of ownership (TCO), increased rack density, and improved power efficiency.
- Consolidate your workloads and increase infrastructure efficiency on a single **unified block and file storage** platform with consistent management and simplicity at scale across the hybrid cloud.
- Boost the security of your data with integrated **real-time ransomware detection** and a comprehensive defense workflow aligned with the NIST cyber resilience framework.
- **HPE Alletra Storage MP B10000 is now NEBS Level 3 certified** – this includes both B10100 and B10200 models
- **HPE Alletra Storage MP B10000 has been rebranded** – previously HPE GreenLake for Block Storage built on HPE Alletra Storage MP.
- **Easily and cost-effectively scale to meet evolving workload demands with disaggregated storage:** New, multi-node switched models enable you to scale capacity and performance independently for greater efficiency and lower cost.
- **Accelerate mission-critical apps with up to 2X performance boost¹.** New 2, 3, and 4 controller node switched models (with a choice of 16-core or 32-core configurations) – adding to the existing option of 2 core controller node switchless models – enables a leap in performance of up to 2X.
- **Scale without limitation with 2X more capacity².** Support for up to 16 JBOF expansion shelves enables you to start small and scale big – from 15.36 TB to around 5.8 PB³ - without disruption.

Overview

- **Maximize efficiency and cost savings with granular capacity upgrades:** Support for 8 to 24 SSDs per enclosure—with the ability to upgrade drives in 2-drive increments and JBOFs in increments of 1—gives you the flexibility to fine-tune, adapt, and elastically scale storage environments in line with specific workload capacity requirements and evolving business needs.
- **Unlock the full power of NVMe for Ethernet networks with NVMe/TCP:** With new NVMe over TCP host protocol support, you get all the benefits of a low latency, high performance NVMe fabric without committing to a dedicated storage network like Fibre Channel. You now have the flexibility to choose the front-end storage protocol that best fits your workloads and existing network infrastructure thanks to support for Fibre Channel, NVMe-oF/FC, NVMe-oF/TCP and iSCSI.
- **Simplify management with enhanced and extended AI-driven performance reporting:** HPE Alletra Storage MP B10000 is powered by industry leading AIOps for infrastructure, delivered via the Data Services Cloud Console (DSCC) on the HPE GreenLake cloud platform. The latest release of the Data Ops Manager app on DSCC continues to eliminate the guesswork for customers when it comes to storage optimization through enhanced and extended AI-based performance reporting and analytics for improved troubleshooting and insights. Key reporting enhancements include headroom utilization trend analysis by volume set, top volume set hotspots by latency, improved workload drift detection, and resource contention detection. With the latest release of the Data Ops Manager, customers also get improved capacity reporting, capacity savings and efficiency metrics, and new sustainability metrics (including carbon utilization emissions and energy consumption).
- **Protect your organization from the threat of a ransomware attack:** The HPE Alletra Storage MP B10000 provides storage teams with all the security and data protection functionality they need to protect their organization from a ransomware attack. The platform integrates cyber-resilient technologies directly into its architecture to strengthen threat detection, containment, and rapid recovery. HPE Alletra Storage MP B10000 ransomware protection framework provides:
 - Real-time ransomware threat detection
 - Automated response workflows to help contain malicious activity
 - Immutable recovery for secure, tamper-proof restore points
 - SIEM/XDR alerting with volume-level identification of affected data
 - Supported by the HPE Cyber Security Ransomware Guarantee

Learn more: HPE Alletra Storage MP B10000 Ransomware Protection Framework (public):

<https://www.hpe.com/psnow/doc/a50013811enw>

- **Take the risk and complexity out of data migration:** Remove the risk and complexity of switching to a new storage platform with multiple options for quick, non-disruptive data migration.
 - With [HPE Peer Motion](#), you can move data from HPE 3PAR, HPE Primera, and HPE Alletra 9000 storage arrays to HPE Alletra Storage MP B10000—online, non-disruptively, and without complex planning or dependency on extra tools.
 - With [HPE Zerto](#), you get a simple (4-click), scalable, and risk-free migration experience for large, complex virtualized environments—regardless of hardware or storage platform.
 - If you don't have the required in-house IT staff availability, expertise, and tools to ensure a successful migration, [HPE Data Migration Services](#) will get you up and running fast. This simplifies the data migration process and provides a flexible way to migrate an organization's critical data safely with minimal impact to operations. You have a choice of several online and offline options for migrating their data to align with their organization's requirements, budgets, and timelines.

Overview

- **Support for QLC SSDs:** Provides lower cost per TB and greater storage density; ideal for heavy read workloads
- **Support for 10GBase-T:** Enables additional host connectivity for legacy infrastructure
- **Numerous software feature improvements:** Including greater system performance, security enhancements, and expanded peer motion support

Notes:

- ¹Compared to existing 2 node (16-core/32-core) switchless configurations
 - ²Compared to release 3 of HPE Alletra Storage MP B10000
 - ³Fully loaded configuration with 16x JBOFs totalling 384 x 15.36 TB NVMe SSDs.
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Standard Features

Hardware

- HPE Alletra Storage MP B10000 is powered by 2-Node, 3-Node, and 4-Node storage controllers and up to 5.9 PB of all NVMe capacity in a single system.

Simplify management with an intuitive cloud experience on premises

- **Simplify operations and move faster with a cloud operational experience:** Reduce on-premises storage complexity across the lifecycle— from install to upgrade—with an AI-driven cloud operational experience powered by the HPE GreenLake Edge to Cloud Platform.
- **Simplified deployment:** Get started in minutes with streamlined device deployment. Simply rack the infrastructure, plug in the power cords, and connect the network cables. With a few clicks, the new system is configured and available in your fleet, ready to serve data for application workloads.
- **Deploy apps faster:** Automate app deployment with intent-based provisioning. Select the storage tier, workload type, capacity, and protection policy, and let AI-driven intelligence automatically optimize your SLAs by recommending the best-suited system across your fleet for your new workload.
- **Unified storage management:** 100% cloud-managed infrastructure means you can manage, monitor, and protect your global storage environment from a single cloud console that's accessible from any location, on any device - so managing hundreds of systems across geographies is as simple as managing one.
- **Invisible upgrades:** Thanks to SaaS-based delivery, new data services instantly become available to you. Data plane software upgrades are non-disruptive and intelligently matched to a given system.

Run any app – without compromise with a 100% data availability guarantee

- **Get peace of mind with 100% data availability guaranteed:** Built on an AI-driven, disaggregated, no single point of failure platform to guarantee resilient 100% data availability for your mission-critical apps.
 - **Advanced DR and HA:** Meet any recovery-point objective (RPO) and recovery-time objective (RTO) strategy with transparent business continuity and automatic fail-over across multiple sites, and provides simple and efficient hybrid cloud data protection for on-premises and cloud-native workloads.
 - **Accelerate your most demanding applications:** Built on a unique, massively parallel, multi-node, and all-active platform, HPE Alletra Storage MP B10000 consolidates traditional and next-generation mission-critical applications at scale with predictable performance and ultra-low latency.
 - **Industry's most advanced AI-Ops:** Predict and prevent disruptions before they occur across the stack and pinpoint issues between storage and VMs and under-utilized resources. Take the guesswork out of managing data infrastructure with AI-driven recommendations that improve performance, drives higher availability, and optimizes resource utilization and planning.
 - **Transformed support experience:** Eliminate time-consuming, frustrating escalations via predictive support automation and direct access to the experts and resources you need.
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Standard Features

HPE Alletra Storage MP B10100 (switchless)						
		B10120	B10130	B10140	B10140 (4N)	Expansion Shelf (JBOF)
Base Chassis		2U	2U	2U	4U (2x 2U)	2U
Number of Nodes		2	2	2	4	2
CPUs per Node		1	1	1	1	1
CPU core count		8-core	16-core	32-core	32-core	8-core
Cache Memory per Node		256 GB	256 GB	512 GB	512 GB	64 GB (not used as cache)
Maximum Number of slots per Node ¹		4	4	4	4	2
Maximum Host Ports per Node		8-ports	12-ports	12-ports	8-ports	N/A
Fibre Channel Host Ports per Node		0 - 8 ports	0 - 12 ports	0 - 12 ports	0 - 8 ports	N/A
10/25 GbE Host Ports per Node	iSCSI	0-4 ports	0-4 ports	0-4 ports	0-4 ports	N/A
	NVMe/TCP	0-4 ports	0-4 ports	0-4 ports	0-4 ports	
	File	0-2 ports	0-2 ports	0-2 ports	0-2 ports	
Max Number of NVMe SSDs per shelf		24	24	24	24	24
Max number of drive shelves per system		1	3 1 base +2 JBOF	3 1 base +2 JBOF	3 2 base +1 JBOF	N/A
Max Raw Capacity per shelf		737.2 TB/670.6 TiB	737.2 TB/670.6 TiB	737.2 TB/670.6 TiB	737.2 TB/670.6 TiB	737.2 TB/670.6 TiB
Max Effective Capacity ² per shelf		1993.5 TB/1813.1 TiB	1993.5 TB/1813.1 TiB	1993.5 TB/1813.1 TiB	1993.5 TB/1813.1 TiB	1993.5 TB/1813.1 TiB
Max Raw Capacity per System		737.2 TB/670.6 TiB	2211.8 TB/2011.6 TiB	2211.8 TB/2011.6 TiB	2211.8 TB/2011.6 TiB	N/A
Max Effective Capacity ² per System		1993.5 TB/1813.1 TiB	6165.6 TB/5607.6 TiB	6165.6 TB/5607.6 TiB	6165.6 TB/5607.6 TiB	N/A

Standard Features

HPE Alletra Storage MP B10200 (switched)				
		B10230	B10240	Expansion Shelf (JBOF)
Base Chassis		2U	2U	2U
Number of Nodes		2, 3, or 4	2, 3, or 4	2
CPUs per Node		1	1	1
CPU core count		16-core	32-core	8-core
Cache Memory per Node		256 GB	512 GB	64 GB (not used as cache)
Maximum Number of slots per Node ¹		4	4	2
Maximum Host Ports per Node		8-ports	8-ports	N/A
Fibre Channel Host Ports per Node		0 - 8 ports	0 - 8 ports	N/A
10/25 GbE Host Ports per Node	iSCSI NVMe/TCP File	0-4 ports 0-4 ports 0-2 ports	0-4 ports 0-4 ports 0-2 ports	N/A
Max Number of NVMe SSDs per shelf		N/A	N/A	24
Max number of expansion shelves per system		16	16	N/A
Max Raw Capacity per shelf		737.2 TB/670.6 TiB	737.2 TB/670.6 TiB	737.2 TB/670.6 TiB
Max Effective Capacity ² per shelf		1993.5 TB/1813.1 TiB	1993.5 TB/1813.1 TiB	1993.5 TB/1813.1 TiB
Max Raw Capacity per System		5.9 PB/5.24 PiB	5.9 PB/5.24 PiB	N/A
Max Effective Capacity ² per System ³		16.6 PB/16.21 PiB	16.6 PB/16.21 PiB	N/A

Notes:

- ¹Slot 3 and Slot 4 can be used for host connectivity (at least one is required), and RCIP is optional; Slot 1 and Slot 2 are used for expansion
- ²Effective capacity assumes 4:1 data compaction ratio (thin provisioning, deduplication, compression, and copy technologies) in a RAID 6 (10+2) configuration. Note TB vs TiB. Actual ratios will vary based on workload. It also factors in 9% free space headroom to ensure optimal array operation.

Host OS Support

Microsoft® Windows® Server and Microsoft® Hyper-V™ | VMware ESX and ESX™i | Red Hat® Enterprise Linux® | SUSE® Linux Enterprise Server (SLES)

Host OS support varies by connectivity protocol to host. For the latest information on supported operating systems refer to Single Point of Connectivity Knowledge for HPE Storage Products (SPOCK):

<http://www.HPE.com/storage/spock>

Service and Support

Warranty

HPE Alletra Storage MP B10000 has 1-year, parts-only warranty. The warranty on all HPE Alletra Storage MP B10000 Solid State Drives (SSD) is 1 year, parts only, and offers unconditional replacement in case of drive failure, media wear-out, or both. Hewlett Packard Enterprise warrants only that the software media will be free of physical defects for a period of ninety (90) days from delivery. For more information about Hewlett Packard Enterprise's Global Limited Warranty and Technical Support, visit: <http://www.HPE.com/storage/warranty>

HPE Services

No matter where you are in your digital transformation journey, you can count on HPE Services to deliver the expertise you need when, where and how you need it. From planning to deployment, ongoing operations and beyond, our experts can help you realize your digital ambitions.

<https://www.HPE.com/services>

Consulting Services

No matter where you are in your journey to hybrid cloud, experts can help you map out your next steps. From determining what workloads should live where, to handling governance and compliance, to managing costs, our experts can help you optimize your operations.

<https://www.HPE.com/services/consulting>

HPE Managed Services

HPE runs your IT operations, providing services that monitor, operate, and optimize your infrastructure and applications, delivered consistently and globally to give you unified control and let you focus on innovation.

[HPE Managed Services | HPE](#)

Operational services

Optimize your entire IT environment and drive innovation. Manage day-to-day IT operational tasks while freeing up valuable time and resources. Meet service-level targets and business objectives with features designed to drive better business outcomes.

<https://www.HPE.com/services/operational>

HPE Complete Care Service

HPE Complete Care Service is a modular, edge-to-cloud IT environment service designed to help optimize your entire IT environment and achieve agreed-upon IT outcomes and business goals through a personalized experience. All delivered by an assigned team of HPE Services experts. HPE Complete Care Service provides:

- A complete coverage approach -- edge to cloud
- An assigned HPE team
- Modular and fully personalized engagement
- Enhanced Incident Management experience with priority access
- Digitally enabled and AI-driven customer experience

<https://www.HPE.com/services/completercare>

Service and Support

HPE Tech Care service

HPE Tech Care service is the operational support service experience for HPE products. The service goes beyond traditional support by providing access to product specific experts, an AI-driven digital experience, and general technical guidance to not only reduce risk but constantly search for ways to do things better. HPE Tech Care service delivers a customer-centric, AI-driven, and digitally enabled customer experience to move your business forward. HPE Tech Care service is available in three response levels. Basic, which provides 9x5 business hour availability and a 2-hour response time. Essential which provides a 15-minute response time 24x7 for most enterprise level customers, and Critical which includes a 6-hour repair commitment where available and outage management response for severity 1 incidents.

<https://www.HPE.com/services/techcare>

HPE Lifecycle Services

HPE Lifecycle Services provide a variety of options to help maintain your HPE systems and solutions at all stages of the product lifecycle. A few popular examples include:

- Lifecycle Install and Startup Services: Various levels for physical installation and power on, remote access setup, installation and startup, and enhanced installation services with the operating system.
- HPE Firmware Update Analysis Service: Recommendations for firmware revision levels for selected HPE products, taking into account the relevant revision dependencies within your IT environment.
- HPE Firmware Update Implementation Service: Implementation of firmware updates for selected HPE server, storage, and solution products, taking into account the relevant revision dependencies within your IT environment.
- Implementation assistance services: Highly trained technical service specialists to assist you with a variety of activities, ranging from design, implementation, and platform deployment to consolidation, migration, project management, and onsite technical forums.
- HPE Service Credits: Access to prepaid services for flexibility to choose from a variety of specialized service activities, including assessments, performance maintenance reviews, firmware management, professional services, and operational best practices.

Notes: To review the list of Lifecycle Services available for your product go to:

<https://www.HPE.com/services/lifecycle>

For a list of the most frequently purchased services using service credits, see the [HPE Service Credits Menu](#)

Other Related Services from HPE Services:

HPE Installation and Startup Service

Provides for the installation of your HPE hardware according to product specifications including options. The HPE service delivery technician will connect the product to a LAN as appropriate and enable remote support to allow for automatic case creation for hardware failures.

HPE Software Installation and Startup Service

Provides deployment of individual software features, helping to ensure proper installation in your storage environment as well as helping you increase the benefit from your storage investment.

Service and Support

HPE Education Services

Training and certification designed for IT and business professionals across all industries. Broad catalogue of course offerings to expand skills and proficiencies in topics ranging from cloud and cybersecurity to AI and DevOps. Create learning paths to expand proficiency in a specific subject. Schedule training in a way that works best for your business with flexible continuous learning options.

<https://www.HPE.com/services/training>

Defective Media Retention

An option available with HPE Complete Care Service and HPE Tech Care service and applies only to Disk or eligible SSD/Flash Drives replaced by HPE due to malfunction.

Consult your HPE Sales Representative or Authorized Channel Partner of choice for any additional questions and services options.

Parts and Materials

HPE will provide HPE-supported replacement parts and materials necessary to maintain the covered hardware product in operating condition, including parts and materials for available and recommended engineering improvements.

Parts and components that have reached their maximum supported lifetime and/or the maximum usage limitations as set forth in the manufacturer's operating manual, product QuickSpecs, or the technical product data sheet will not be provided, repaired, or replaced as part of these services.

How to Purchase Services

Services are sold by Hewlett Packard Enterprise and Hewlett Packard Enterprise Authorized Service Partners:

- Services for customers purchasing from HPE or an enterprise reseller are quoted using HPE order configuration tools.
- Customers purchasing from a commercial reseller can find services at <https://ssc.HPE.com/portal/site/ssc/>

AI Powered and Digitally Enabled Support Experience

Achieve faster time to resolution with access to product-specific resources and expertise through a digital and data driven customer experience

Sign into the HPE Support Center experience, featuring streamlined self-serve case creation and management capabilities with inline knowledge recommendations. You will also find personalized task alerts and powerful troubleshooting support through an intelligent virtual agent with seamless transition when needed to a live support agent.

<https://support.HPE.com/hpesc/public/home/signin>

To learn more about HPE Services, please contact your Hewlett Packard Enterprise sales representative or Hewlett Packard Enterprise Authorized Channel Partner. Contact information for a representative in your area can be found at "Contact HPE" <https://www.HPE.com/us/en/contact-HPE.html>

For more information

<http://www.HPE.com/services>

Service and Support

Consume IT On Your Terms

[HPE GreenLake Solutions for HPE Alletra Storage MP B10000](#)

HPE GreenLake Solutions with HPE Alletra Storage MP B10000 are available for Customers who wish to pursue Storage-as-a-Service (SaaS) or any Hybrid Cloud solutions based on pay-per-use economics. Whether customers require pre-configured or customized solutions and services, or they simply want to pay only for the data they write to disk via a simple monthly payment – HPE Alletra Storage MP B10000 is readily available for customers looking to procure Storage-as-a-Service (SaaS) in both simple, standalone or complex GreenLake solutions.

For More Information, please visit [HPE Alletra Storage MP B10000](#)

HPE Timeless Storage

The HPE Timeless Storage Program that allows customers to refresh their existing technology via non-disruptive controller upgrades is now available on HPE Alletra Storage MP B10000. Some Key features of the program include:

- No forklift upgrades or data migration required.
- Customers shall enjoy existing guarantees including 100% Data Availability, StoreMore (data efficiency), and Satisfaction guarantees.
- The program will be launched on HPE Alletra Storage MP B10000, with no change to sales motion, support model, etc.
- The program is available when sold as part of:
 - HPE Alletra Storage MP B10000 or
 - HPE Green Lake for Block Storage Private Cloud Business Edition
- HPE Timeless Storage
 - HPE Alletra Storage MP B10030 Tech Refresh Single Node - S5E84A
 - HPE Alletra Storage MP B10040 Tech Refresh Single Node - S5E85A

For more details on the program please refer to the datasheet:

[HPE Timeless Storage – For HPE Alletra Storage MP B10000](#)

Configuration Information

Step 1: Choose the storage chassis and controller nodes

HPE Alletra Storage MP B10000 configuration starts with the selection of the Chassis and Controller Nodes. The Chassis is a 2U 2-node chassis that includes 24 bays for small form factor NVMe SSD drives for switchless configuration. Switched configurations do not allow for SSD drives in the same chassis as the controller nodes—NVMe SSDs are added via expansion shelves (JBOFs). The Chassis does not include any controller nodes or power supplies—these must be added.

HPE Alletra Storage MP 10000 Chassis

	SKU
HPE Alletra Storage MP B10100 Base Configuration	SOB84A
HPE Alletra Storage MP B10200 Base Configuration	S1R06A
– One (1) SOB84A (switchless) or S1R06A (switched) SKU must be ordered for each configuration	
HPE Alletra Storage MP 10000 2U Chassis	R7C75A
HPE Alletra Storage MP 10000 2U TAA-compliant Chassis	SOU32A
– One (1) Alletra Storage MP 10000 Chassis SKU must be ordered for each switchless array.	
– One (1) or two (2) Alletra Storage MP Chassis 10000 SKUs must be ordered for each switched array	
– For switchless configurations, the Alletra Storage MP 10000 Chassis can host two controller nodes and up to 24 small form factor NVMe drives in a 2U chassis.	
– For switched configurations, the Alletra Storage MP 10000 Chassis can host two controller nodes in a 2U chassis.	
– The Alletra Storage MP 10000 Chassis does not include any controller nodes or Power Supplies.	

HPE Alletra Storage MP B10000 Controller Nodes

Switchless controller nodes

	SKU
HPE Alletra Storage MP B10120 Controller Node	SOS38A
HPE Alletra Storage MP B10120 TAA-compliant Controller Node	SOS40A
HPE Alletra Storage MP B10130 Controller Node	SOR21A
HPE Alletra Storage MP B10130 TAA-compliant Controller Node	SOS41A
HPE Alletra Storage MP B10140 Controller Node	S4V66A
HPE Alletra Storage MP B10140 TAA-compliant Controller Node	S4V67A

Switched controller nodes

HPE Alletra Storage MP B10230 Controller Node	R7D02A
HPE Alletra Storage MP B10230 TAA-compliant Controller Node	SOS42A
HPE Alletra Storage MP B10240 Controller Node	R7D03A
HPE Alletra Storage MP B10240 TAA-compliant Controller Node	SOR20A
– Each Controller Node SKU includes one controller node	
– Two (2) Controller Node SKUs must be ordered per chassis	
- For switched configurations, the optional second chassis must have one (1) or two (2) controller nodes. Switched configurations support 2, 3, or 4 controller nodes - The HPE Alletra Storage MP B10140 (4N) switchless 4-Node configuration requires two (2) chassis, both populated with two (2) B10140 (32c) controller nodes	
– Controller nodes need to be of the same type. Different controller node types cannot be mixed in the same chassis	
– Each switched B10200 Controller Node contains two (2) OCP slots for host adapters (Slot 3 and Slot 4), and Slot 1 and Slot 2 are used for expansion	

Configuration Information

- For a switchless system, each B10140 100 GbE Controller Node (SOS39A or SOS43A) requires (1) 100 GbE 2-port OCP HBA (S2A68A) in Slot 1 (see Adapters section); maximum of two (2) HBAs for host connectivity
- For switchless systems, B10130 (SOR21A, SOR41A) and B10140 (S4V66A, S4V67A) each Controller Node can support up to three (3) HBAs for host connectivity
- When Expansion Shelves are configured on a switchless system, each B10130 or B10140 Controller Node requires (1) 100 GbE 2-port OCP HBA (S2A68A) in Slot 2 (see Adapters section)
- For a switched system, each Controller Node requires (1) 100 GbE 2-port OCP HBA (S2A68A) in Slot 1 and (1) 100 GbE 2-port OCP HBA (S2A68A) in Slot 2 (see Adapters section)

HPE Alletra Storage MP Switches

	SKU
HPE Alletra Storage MP 32-port 100 GbE Switch Bundle	S1R08A
HPE Alletra Storage MP 32-port 100 GbE -48VDC Switch Bundle	S1S69A
HPE Switch Pair Installation Kit with 4U Cable Tray	SOA95A
HPE Switch Pair Installation Kit without Tray	S3V78A
HPE Alletra Storage 2U Cable Tray	S4Q37A
– When switched systems are configured, min two (2)/max two (2) of the switch bundles are added to the configuration – The Alletra Storage MP switch bundles can only be used for interconnectivity of the Alletra Storage MP B10000 controller nodes and the Alletra Storage MP 10000 expansion shelf nodes – The Alletra Storage MP switch bundles cannot be used for connectivity to host servers or any other systems – The Alletra Storage MP -48VDC switch bundles (S1S69A) are used when DC power is selected for the system – Switched configurations require an Installation Kit— • The 4U Cable Tray Installation Kit (SOA95A) is the default for all switched configurations and includes a 4U cable tray and the installation kit; this kit is required if more than 8 chassis are configured (controller chassis plus JBOF chassis) • The 2U Cable Tray (S4Q37A) can be used if 8 or fewer chassis are configured (controller chassis plus JBOF chassis). The Installation Kit (S3V78A) must also be ordered when ordering the 2U Cable Tray • The Installation Kit without a tray (S3V78A) can be selected for switched configurations that will not grow beyond a total four (4) chassis (controller chassis plus JBOF chassis)	

Step 2: Choose power supplies

Description	SKU
HPE Alletra Storage MP C14 1600 W AC Power Supply	R7C76A
HPE Alletra Storage MP C14 2200 W AC Power Supply	R9Z97A
HPE Alletra Storage MP 1600 W -48VDC Power Supply	R7C78A
HPE Alletra Storage MP C20 1500 W /1800 W AC Power Supply	R7C77A
– The Alletra Storage MP 10000 Chassis requires two (2) Power Supplies per chassis – Power Supplies are factory integrated in the Storage chassis for shipment – The C14 1600 W AC Power Supply (R7C76A) is 80 PLUS Platinum certified – The C14 1600 W AC Power Supply (R7C76A) supports high-line; no low-line support – The C14 2200 W AC Power Supply (R9Z97A) is 80 PLUS Titanium certified for EU Lot9 requirements – The C20 1500 W /1800 W AC Power Supply (R7C77A) is 80 PLUS Titanium certified for EU Lot9 requirements – The C14 2200 W AC Power Supply (R9Z97A) supports high-line; no low-line support – The C20 1500 W /1800 W AC Power Supply (R7C77A) supports both low-line and high-line (switching)	

Configuration Information

- The C20 1500 W AC Power Supply (R7C77A) requires a power cord with a C19 connector
- The DC Power Supply (R7C78A) not available for China, Taiwan, and India
- R9Z97A and R7C77A are not NEBS Level 3 certified at this time

HPE Alletra Storage MP Power Cord

	SKU
HPE C13 - C14 250 V 10Amp Black 1.4 m WW Power Cord	R9R52A
HPE C13 - C14 250 V 10Amp Gray 1.4 m WW Power Cord	R9S00A
HPE C13 - C14 250 V 10Amp Black 1.37 m IN Power Cord	R9R53A
HPE C13 - C14 250 V 10Amp Gray 1.37 m IN Power Cord	R9S01A
HPE C13 - C14 250 V 10Amp Black 2 m WW Power Cord	S3L68A
HPE C13 - C14 250 V 10Amp Gray 2 m WW Power Cord	S3L69A
HPE C13 - C14 250 V 10Amp Black 2 m IN Power Cord	S3L70A
HPE C13 - C14 250 V 10Amp Gray 2 m IN Power Cord	S3L71A
HPE C19 - NEMA 5-15P 125 V 15Amp US Power Cord	R9R71A
HPE NEMA L6-30P - C19 250 V 20Amp 4.5 m US/JP Power Cord	R9R74A
HPE C19 - JIS8303 125 V 16Amp TW/JP Power Cord	R9R73A
HPE C19 - C20 250 V 16Amp Black 2.5 m WW Power Cord	R9S02A
HPE C19 - C20 250 V 16Amp Gray 2 m WW Power Cord	R9R66A
HPE C19 - C20 250 V 16Amp Black 2.5 m IN Power Cord	R9S03A
HPE C19 - C20 250 V 16Amp Gray 2 m IN Power Cord	R9R67A
HPE 1600 W -48VDC 600 V 3.5 m Power Cable Kit	P22173-B21
HPE 1600 W -48VDC Power Cable Lug Kit	P36877-B21
– R9R52A, R9S00A, R9S02A, and R9R66A are for worldwide use except for India – R9R53A, R9S01A, R9S03A, and R9R67A are for India use only – The 2 m power cords are required for the switches	

Step 3: Choose adapters

Host adapters are used for connection to hosts. They can be ordered standalone to be installed in the field or they can be factory integrated into controller nodes. HPE Alletra Storage MP B10000 does not have any built-in host ports therefore any configuration needs to have at least one (1) host adapter per controller node. For RCIP remote copy, the Replication HBAs ([R7C82A](#)) can be ordered as an option.

HPE Alletra Storage MP Host Adapters

	SKU
HPE Alletra Storage MP 32 GB 4-port Fibre Channel OCP LPm37004 Host Bus Adapter	S2S64A
HPE Alletra Storage MP 32/64 GB 4-port Fibre Channel Host Bus Adapter	R7C90A
HPE Alletra Storage MP 10/25 GbE 4-port Host Bus Adapter	R7C82A

HPE Alletra Storage MP Replication

HPE Alletra Storage MP 10/25 GbE 4-port Host Bus Adapter	R7C82A
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HPE Alletra Storage MP Backend Connectivity

HPE Alletra Storage MP 100 GbE 2-port OCP Host Bus Adapter	S2A68A
HPE 100 GB QSFP28 to QSFP28 0.5 m Direct Attach Copper Cable	R8M59A
HPE Aruba Networking 100G QSFP28 to QSFP28 1 m Direct Attach Copper Cable	R9F77A
HPE Aruba Networking 100G QSFP28 to QSFP28 2 m Active Optical Cable	R9F76A

Configuration Information

Requirements for 100 GbE backend connectivity

The table below shows the adapter and cable requirements for backend connectivity for both switched and switchless configurations. The recommended cables are shown, alternate active optical cable (AOC) lengths in Step 5 may be used.

Alletra Storage MP B10100 Nodes (switchless)	100 GbE HBA per node	Cables per node
B10120 controller nodes	N/A	N/A
B10130 controller nodes	(1) S2A68A if JBOF used	(2) R9F77A if JBOF used
B10140 controller nodes	(1) S2A68A if JBOF used	(2) R9F77A if JBOF used
B10140 100 GB controller nodes	(1) S2A68A and (1) S2A68A if JBOF used	(1) R8M59A and (2) R9F77A if JBOF used
B10010 expansion shelf nodes (JBOF)	(1) S2A68A	N/A

Alletra Storage MP B10200 Nodes (switched)	100 GbE HBA per node	Cables per node
B10230 controller nodes	(2) S2A68A	(4) R9F76A
B10240 controller nodes	(2) S2A68A	(4) R9F76A
B10010 expansion shelf nodes (JBOF)	(1) S2A68A	(2) R9F76A (Perf Optimized) (1) R9F76A (Cap Optimized)

Host Bus Adapter Transceiver Kit

	SKU
HPE 10 GB SFP+ Short Wave 1-pack Pull Tab Optical Transceiver	Q2P65A
HPE 10GBASE-T SFP+ RJ45 30 m 1-pack Transceiver	R0R41B
HPE Aruba Networking 10G SFP+ LC LR 10 km SMF Transceiver	J9151E
HPE 25 GB SFP28 Short Wave Extended Temperature 1-pack Pull Tab Optical Transceiver	Q2P64B
HPE Aruba Networking 25G SFP28 LC LR 10 km SMF Transceiver	JL486A
HPE 32 GB SFP28 Short Wave 1-pack LP Pull Tab Optical Transceiver	S3N85A
HPE 32 GB SFP28 Short Wave 1-pack Pull Tab Optical Transceiver	Q2P62A

Notes:

- Each Controller Node must have at least one Fibre Channel HBA (S2S64A), or one 10/25 GbE HBA (R7C82A) for host connectivity. A Controller Node without any host adapters is not a supported configuration.
- Maximum of two (2) or three (3) Fibre Channel HBA (S2S64A) can be ordered per Controller Node. Mixing of different Fibre Channel HBA SKUs on the same system is not supported
- Maximum of two (2) 10/25 GbE HBA (R7C82A) can be ordered per Controller Node.
- For 2-node switchless systems, B10130 (SOR21A, SOR41A) and B10140 (S4V66A, S4V67A) each Controller Node can support up to three (3) HBAs for host connectivity—this can be a mix of FC and 10/25 GbE HBAs.
- Refer to SPOCK and the Alletra Storage MP iSCSI Best Practices document on Seismic for iSCSI configuration details.
- For Unified Block and File, the 10/25 GbE HBA (R7C82A) must be installed in Slot 4 with Ports 1 and 2 dedicated to File. Unified Block and File requires ArcusOS 10.5.0 or later
- The S2S64A 32 GB Fibre Channel Adapter includes two (2) 32 GB transceivers. Additional S3N85A 32 GB transceivers are optional and must be ordered in pairs.
- The R7C90A 32/64 GB Fibre Channel Adapter does not include any transceivers. The Q2P62A 32 GB transceivers must be ordered in pairs separately—min 2, max 4 per HBA

Configuration Information

- Both the R7C90A and S2S64A Fibre Channel HBAs are 64 GB capable; currently only 32 GB SFPs are supported
- The 10/25 GbE Adapter does not include any transceivers, the transceivers must be ordered in pairs separately.
- For each 10/25 GbE Adapter, the 10 GB SFPs or 25 GB SFPs can be ordered with a minimum two (2), and maximum four (4).
- A pair of 10 GB SFPs and a pair of 25 GB SFPs can be mixed in the same 10/25 GbE Adapter.
- 10/25 GB SFPs are not required if DAC cables are selected
- The 25 GB SFP supports only the 25 GB speed—it does not support 10 GB or 1 GB
- The 10 GB SFP supports only the 10 GB speed—it does not support 25 GB or 1 GB
- S2S64A and S3N85A are not NEBS Level 3 certified at this time

Step 4: Choose drives

Drives are orderable at the time the array is purchased and can be added in the future when additional capacity is required. HPE Alletra Storage MP B10000 supports SSDs with NVMe interface and QLC or TLC NAND technology.

HPE Alletra Storage MP Encrypted TLC Drives

	SKU
HPE Alletra Storage MP 1.92 TB NVMe SFF Self-encrypting SSD	R9H66A
HPE Alletra Storage MP 3.84 TB NVMe SFF Self-encrypting SSD	R9H67A
HPE Alletra Storage MP 7.68 TB NVMe SFF Self-encrypting SSD	R9H68A
HPE Alletra Storage MP 15.36 TB NVMe SFF Self-encrypting SSD	R9H69A
HPE Alletra Storage MP 30.72 TB NVMe SFF Self-encrypting SSD	S1Z68A
HPE Alletra Storage MP 1.92 TB NVMe SFF FIPS Encrypted SSD	R9H70A
HPE Alletra Storage MP 3.84 TB NVMe SFF FIPS Encrypted SSD	R9H71A
HPE Alletra Storage MP 7.68 TB NVMe SFF FIPS Encrypted SSD	R9H72A
HPE Alletra Storage MP 15.36 TB NVMe SFF FIPS Encrypted SSD	R9H73A
HPE Alletra Storage MP 30.72 TB NVMe SFF FIPS Encrypted SSD	S4P49A
HPE Alletra Storage MP 1.92 TB NVMe SFF FIPS Encrypted TAA-compliant SSD	SOQ01A
HPE Alletra Storage MP 3.84 TB NVMe SFF FIPS Encrypted TAA-compliant SSD	SOQ02A
HPE Alletra Storage MP 7.68 TB NVMe SFF FIPS Encrypted TAA-compliant SSD	SOQ03A
HPE Alletra Storage MP 15.36 TB NVMe SFF FIPS Encrypted TAA-compliant SSD	SOQ04A
HPE Alletra Storage MP 7.68 TB NVMe SFF FIPS Encrypted Optional SSD	S7J58A
HPE Alletra Storage MP 15.36 TB NVMe SFF FIPS Encrypted Optional SSD	S7J59A
HPE Alletra Storage MP 7.68 TB NVMe SFF Self-encrypting Optional SSD	S7L76A
HPE Alletra Storage MP 15.36 TB NVMe SFF Self-encrypting Optional SSD	S7L77A

Notes:

- Minimum of eight (8) TLC SSDs per system
- Minimum of eight (8) TLC SSDs per chassis
- Mixing of different capacity SSDs in single system is not supported
- Mixing of TLC and QLC SSDs in a single system is not supported

HPE Alletra Storage MP Encrypted QLC Drives

	SKU
HPE Alletra Storage MP 15.36 TB NVMe SFF QLC SSD	S3L73A
HPE Alletra Storage MP 30.72 TB NVMe SFF QLC SSD	S3L74A

Notes:

- Minimum of eight (8) QLC SSDs per system
- Minimum of eight (8) QLC SSDs per chassis

Configuration Information

- Mixing of different capacity SSDs in single system is not supported
- Mixing of TLC and QLC SSDs in a single system is not supported
- QLC drives are self-encrypting
- When 30 TB QLC SSDs are used, system operating temperature is 5° to 30° C
- Always reference sizing tools (e.g., Ninja Online) when using QLC SSDs to ensure the QLC drive performance and wear life is appropriate for the given workload
- QLC SSDs have a lower Drive Writes Per Day (DWPD) endurance compared to TLC SSDs. DWPD indicates the number of times the entire drive can be overwritten each day for its entire life. The QLC SSDs in the HPE Alletra Storage MP have a DWPD of 0.26 whereas the TLC SSDs have a DWPD of 1.0.
- S1Z68A, S3L73A and S3L74A are not NEBS Level 3 certified at this time

Step 5: NVMe Expansion Shelf Just a Bunch of Flash (JBOF)

The HPE Alletra Storage MP 10000 NVMe all-flash expansion shelf (JBOF) is a 2U chassis to add additional system capacity. The expansion shelf supports 8-24 NVMe SSDs, in two-drive increments with up to two (2) Expansion Shelves per system (switchless) and up to sixteen (16) Expansion Shelves per system (switched). Each expansion shelf requires:

Notes:

- Min 2/Max 2 of Expansion Shelf Nodes
- Min 2/Max 2 Power Supplies and power cords
- Min 8/Max 24 SSDs (see SSDs listed above)
- Min 1/Max 1 100 GbE 2-port OCP HBA (S2A68A) per Expansion Shelf Node
- Switchless: Min 2/Max 2 100G cables (1 per Expansion Shelf Node), 100G 1 m DAC Cable (R9F77A) recommended—Active Optical Cables listed also supported
- Switched: Min 2/Max 2 of 100G Active Optical Cable (AOC) for Performance Optimized configurations (1-8 JBOFs); the R9F76A 100G QSFP28 to QSFP28 2 m Active Optical Cable is the default; other AOC lengths listed also supported.
- Switched: Min 1/Max 1 of 100G Active Optical Cable (AOC) for Capacity Optimized configurations (9-16 JBOFs); the R9F76A 100G QSFP28 to QSFP28 2 m Active Optical Cable is the default; other AOC lengths listed also supported.

Description

	SKU
HPE Alletra Storage MP 10001 NVMe Configure-to-order Expansion Shelf	S1J10A
HPE Alletra Storage MP 10010 Expansion Shelf Node	S1R28A
HPE Alletra Storage MP 10001 NVMe Configure-to-order TAA-compliant Expansion Shelf	SOS46A
HPE Alletra Storage MP 10010 TAA-compliant Expansion Shelf Node	S1R29A
HPE Alletra Storage MP 100 GbE 2-port OCP Host Bus Adapter	S2A68A
HPE Aruba Networking 100G QSFP28 to QSFP28 1 m Direct Attach Copper Cable	R9F77A
HPE Aruba Networking 100G QSFP28 to QSFP28 2 m Active Optical Cable	R9F76A
HPE Aruba Networking 100G QSFP28 to QSFP28 7 m Active Optical Cable	R9F79A
HPE Aruba Networking 100G QSFP28 to QSFP28 15 m Active Optical Cable	R9F80A
HPE Aruba Networking 100G QSFP28 to QSFP28 30 m Active Optical Cable	R9F81A
HPE Storage 100 GbE QSFP28 to QSFP28 5 m Extended Temperature Active Optical Cable	S2T38A
HPE Storage 100 GbE QSFP28 to QSFP28 10 m Extended Temperature Active Optical Cable	S2T39A

Configuration Information

Description

SKU

HPE Storage 100 GbE QSFP28 to QSFP28 15 m Extended Temperature Active Optical Cable

S2T40A

HPE Storage 100 GbE QSFP28 to QSFP28 25 m Extended Temperature Active Optical Cable

S2T41A

- The minimum supported quantity is eight (8) drives per chassis
- Drives can be ordered in increments of two drives from 8 to 24 drives per chassis
- HPE Alletra Storage B10000 supports only RAID 6
- For initial configurations, and within a system (base array plus any expansion shelves), all drives must be of the same capacity and type. Mixing drives of different capacities in the same system is not allowed.
- Drives must be loaded starting from the leftmost slot (slot 1) to the right and leaving no empty slots between drives.
- The switchless 2-node 16-core and 32-core systems support up to two (2) expansion shelves
- The switchless 4-node 32-core systems support up to one (1) expansion shelf
- The switched systems support up to eight (8) expansion shelves (Performance Optimized), or 9-16 expansion shelves (Capacity Optimized).
 - When a switched system is ordered with 9-16 JBOFs, it is configured as Capacity Optimized
 - When a switched system is ordered with 1-8 JBOFs, it is configured as Performance Optimized
- Capacity Optimized systems require HPE Alletra Storage MP ArcusOS version 10.4.0 or later
- Capacity Optimized configurations are not supported with the 30.72 TB QLC drives
- The 8-core systems do not support expansion shelves
- When expansion shelves are used, drive quantities should be balanced across all shelves—base array shelf and expansion shelves

HPE Encryption License

SKU

HPE Storage FIPS Data Encryption LTU

SOA99A

HPE Storage FIPS Data Encryption E-LTU

SOA99AEE

HPE Storage Data Encryption LTU

SOA98A

HPE Storage Data Encryption E-LTU

SOA98AEE

- A data encryption license (LTU) is required to enable encryption on HPE Alletra Storage MP B10000. One encryption license is required for each encrypted storage solution. Once encryption is enabled on the storage solution, it cannot be disabled.
- Encryption can be turned on, non-disruptively, at any time, even after data has been written to the system.
- Supports Utimaco® Enterprise Secure Key Manager (ESKM) 4.0, 5.0 and Gemalto® SafeNet KeySecure k460 centralized key management
- The local key manager is included in the HPE Alletra Storage ArcusOS. There is not a separately orderable part number for the local key manager

Step 6: Choose cables for host connectivity

HPE Alletra Storage MP B10000 requires cables for host connectivity, and replication. The OM4 fiber cables are used for host connectivity and Peer Motion.

OM4 Cables

SKU

HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 1 m Cable

QK732A

HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 2 m Cable

QK733A

Configuration Information

Description

	SKU
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 5 m Cable	QK734A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 15 m Cable	QK735A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 30 m Cable	QK736A
HPE Premier Flex LC/LC Multi-mode OM4 2 Fiber 50 m Cable	QK737A

Direct Attach Copper Cables (10 GbE) - HPE Networking Comware

	SKU
HPE Networking X240 10G SFP+ SFP+ 3 m DAC Cable	JD097C
HPE Networking X240 10G SFP+ SFP+ 5 m DAC Cable	JG081C
HPE Networking Comware X240 40G QSFP+ to 4x10G SFP+ 3 m Direct Attach Copper Splitter Cable	JG330A
HPE Networking Comware X240 40G QSFP+ to 4x10G SFP+ 5 m Direct Attach Copper Splitter Cable	JG331A

Direct Attach Copper Cables (10 GbE) – HPE Aruba Networking

HPE Aruba Networking 10G SFP+ to SFP+ 3 m Direct Attach Copper Cable	J9283D
HPE Aruba Networking 10G SFP+ to SFP+ 7 m Direct Attach Copper Cable	J9285D

Direct Attach Copper Cables (10 GbE) – Cisco

HPE C-series 3 m Passive Copper SFP+ Cable	K2Q21A
HPE C-series 5 m Passive Copper SFP+ Cable	K2Q22A
HPE C-series SFP+ to SFP+ Active Copper 7.0 m Direct Attach Cable	QK701A

Direct Attach Copper Cables (10 GbE) - HPE BladeSystem

HPE BladeSystem c-Class 10 GbE SFP+ to SFP+ 3 m Direct Attach Copper Cable	487655-B21
HPE BladeSystem c-Class 10 GbE SFP+ to SFP+ 5 m Direct Attach Copper Cable	537963-B21

Smart Active Optical Cables (25 GbE) - HPE

HPE 25 GbE SFP28 to SFP28 3 m Smart Active Optical Cable	Q9S67A
HPE 25 GbE SFP28 to SFP28 5 m Smart Active Optical Cable	Q9S68A

AOC/DAC Cables– HPE

HPE 25 GB SFP28 to SFP28 3 m Direct Attach Copper Cable	844477-B21
HPE 25 GB SFP28 to SFP28 7 m Active Optical Cable	844483-B21
HPE 100 GB QSFP28 to 4x25Gb SFP28 3 m Direct Attach Copper Cable	845416-B21

Ethernet Cables (Management)

HPE RJ45 to RJ45 Cat5e Black M/M 7.6ft 1-pack Data Cable	C7535A
HPE 4.3 m /14ft CAT5 RJ45 M/M Ethernet Cable	C7536A
HPE 7.6 m /25ft CAT5 RJ45 M/M Ethernet Cable	C7537A
HPE 15.2 m /50ft CAT5 RJ45 M/M Ethernet Cable	C7542A

Notes: For the latest information refer to Single Point of Connectivity Knowledge for HPE Storage Products (SPOCK): <http://www.HPE.com/storage/spock>

Configuration Information

Step 7: Choose bluetooth connectivity kit option

HPE Alletra Storage MP 10000 supports a Chassis Discovery Module (CDM) which enables the system to have its setup initiated through a Bluetooth connection to a mobile app. This is an optional feature which requires the Bluetooth Connectivity Kit to be included.

Description

HPE Alletra Storage MP CDM Bluetooth Connect Kit

SKU

S0U79A

Step 8: Choose racking options

HPE Alletra Storage MP B10000 is compatible with most industry standard 4-post EIA 19-inch racks with square mounting holes. The B10000 can be factory configured and shipped in a rack or shipped without a rack for field integration into an existing rack. The racks used for factory integration are the HPE G2 Advanced Series Racks or the HPE G2 Enterprise Series Racks.

Factory Integration

A factory integrated HPE Alletra Storage MP B10000 is configured into the HPE Intelligent Series Rack with the appropriate power distribution units (PDUs). Other products such as servers or back-up products can be factory integrated in the rack and different PDUs can be added or selected (if needed) only via HPE Factory Express Services. Multiple HPE B10000 arrays can be factory integrated in the same rack.

HPE Intelligent Series Racks

	SKU
HPE 42U 600mmx1200mm G2 Enterprise Shock Rack	P9K40A
HPE G2 Rack 42U 1200 mm Side Panel Kit	P9L16A
HPE 42U 600mmx1200mm G2 Kitted Advanced Shock Rack with Side Panels and Baying	P9K10A
HPE 42U 600mmx1075mm G2 Enterprise Shock Rack	P9K38A
HPE G2 Rack 42U 1075 mm Side Panel Kit	P9L15A
HPE 42U 600mmx1075mm G2 Kitted Advanced Shock Rack with Side Panels and Baying	P9K08A
– HPE recommends using HPE Alletra Storage MP B10000 in 1200 mm deep racks with 3-phase power because this combination provides the best rack density.	

Notes:

- For more information on rack options: [Rack and Power Infrastructure Products](#)
- For more information on PDUs: <https://www.HPE.com/us/en/product-catalog/servers/power-distribution-units.html>

Step 9: Choose software

Hewlett Packard Enterprise provides an extensive selection of features in the HPE Alletra Storage ArcusOS software, including Virtual Copy, Remote Copy, Priority Optimization, Peer Motion, and Peer Persistence, etc. Data Encryption is offered separately with a LTU (see the Drive section for more information).

HPE Alletra Software and Support SaaS

HPE Alletra Storage MP B10000 include a subscription to HPE Alletra Storage MP B10000 Software and Support SaaS that includes all-inclusive software features for the specified raw capacity and term, and it enables cloud-based management of the array from the HPE Data Services Cloud Console and access to data services, and related support.

Configuration Information

- HPE Alletra Storage MP B10000 Software and Support SaaS SKU must be selected for each storage solution
- The quantity of SaaS is determined by the raw capacity, in TB, configured. For example, if 100 TB is configured, then the SaaS quantity will equal 100 for the term ordered
- The standard terms are shown below; other terms may be available by exception

HPE Alletra Software and Support SaaS

	SKU
HPE Alletra Storage MP B10000 per TB 3-year Software and Support SaaS	S3Q00AAE
HPE Alletra Storage MP B10000 per TB 4-year Software and Support SaaS	S3Q01AAE
HPE Alletra Storage MP B10000 per TB 5-year Software and Support SaaS	S3Q02AAE

HPE Alletra Software and Support SaaS Upgrades and Renewal

	SKU
HPE Alletra Storage MP B10000 per TB GLP Upgrade 1-year Software and Support SaaS	S6M49AAE
HPE Alletra Storage MP B10000 per TB GLP Upgrade 3-year Software and Support SaaS	S6M50AAE
HPE Alletra Storage MP B10000 per TB GLP Upgrade 4-year Software and Support SaaS	S6M51AAE
HPE Alletra Storage MP B10000 per TB GLP Upgrade 5-year Software and Support SaaS	S6M52AAE
HPE Alletra Storage MP B10000 per TB GLP Upgrade 6-year Software and Support SaaS	S6M53AAE
HPE Alletra Storage MP B10000 per TB GLP Upgrade 7-year Software and Support SaaS	S6M54AAE
HPE Alletra Storage MP B10000 per TB Renewal Software and Support SaaS	S6J58AAE

In addition to the HPE Alletra Software and Support SaaS SKU, each system will have an HPE Alletra Storage ArcusOS LTU SKU. The quantity for the LTU SKU will always equal one (1) and will match the term of the SaaS SKU and Support

HPE Alletra Storage ArcusOS per TB 3-year LTU	S3Q00AA
HPE Alletra Storage ArcusOS per TB 4-year LTU	S3Q01AA
HPE Alletra Storage ArcusOS per TB 5-year LTU	S3Q02AA

HPE Alletra Storage MP Disconnected

HPE Alletra Storage MP Disconnected is a simple-to-manage, fully disconnected, and secure block storage solution for an isolated environment without internet connections. This solution is built on HPE Alletra Storage MP B10000 with data services delivered on the HPE GreenLake Dedicated Platform for regulated environments, the disconnected counterpart to the HPE GreenLake Cloud platform. It offers the same cloud experience as HPE GreenLake Cloud, but within your data centers in an isolated environment. It allows you to meet regulatory requirements with ironclad, highly available enterprise-grade block storage, leveraging a cloud-like operational experience to efficiently manage your data in dark sites.

New purchase of B10000 Array

For ordering, in the disconnected menu select the SKU S4W12A – HPE Alletra Storage MP Disconnected. This will populate the appliance, and all other components associated with On-Premises management. HPE Alletra Storage MP Disconnected Solution Add-on Tracking (S6C37A) is used when adding a new HPE Alletra Storage MP B10000 to an existing HPE GreenLake Dedicated Platform.

Description	SKU
HPE Alletra Storage MP Disconnected LTU	S4W12A
HPE Alletra Storage MP Disconnected Solution Add-on Tracking	S6C37A

Configuration Information

Existing B10000 customer on air-gapped/cloud connected license

This solution is intended for customers who already have an HPE Alletra MP B10000 system and wish to convert it from a cloud-connected or air-gapped configuration to a fully disconnected deployment.

To begin the conversion process, users should search for “Disconnected” or “HPE Alletra Storage MP Disconnected” in the Search Product field and select SKU ‘S4W12A – HPE Alletra Storage MP Disconnected’. Upon selecting the create option, the user will be guided to the Solution Wizard. Within the wizard, the solution type must be set to ‘Expansion Solution (Brownfield)’, which will automatically apply default selections including a TAA-compliant server and a 5-year support term. Storage selection is not required, as this configuration is applicable only to existing B10000 systems. The resulting Bill of Materials (BOM) will include the disconnected appliance and the necessary licenses to complete the conversion, specifically SKU S4W12AAE – HPE Alletra Storage MP B10000 Disconnected E-LTU.

Description

HPE Alletra Storage MP Disconnected E-LTU

SKU

S4W12AAE

Step 10: Choose support

Choose HPE Services Tech Care to experience the new operational service for HPE products. For Alletra Storage MP B10000, HPE Services Tech Care is available in three response levels: Basic which provides next-business-day parts exchange where available, Essential which provides 4-hour parts exchange where available, and Critical which includes a 6-hour Hardware-Call-to-Repair for severity 1 and severity 2 incidents where available and outage management response for severity 1 incidents.

All levels of Tech Care enjoy 7x24 access to experienced technical engineers upon calling HPE for support.

<https://www.HPE.com/psnow/doc/a50003571enw>

Step 11 – Choose installation options

Storage Installation and Startup Service

HPE Alletra Storage MP B10000 Installation and Startup Service provides deployment of your B10000, helping to ensure proper installation in your storage environment as well as helping you increase the benefit from your storage investment. The service provides activities required to help you deploy your HPE Alletra Storage MP B10000 into operation.

Self-Installation

Customers and partners also have the option to self-install HPE Alletra Storage MP B10100 (switchless). The self-installation option can be selected in the configurator tool (OCA) and will remove the Storage Installation and Startup Service from the quote. HPE Alletra Storage MP B10200 (switched) models and configurations are not eligible for self-installation at this time.

To successfully install the HPE Alletra B10000 Storage, the installer should:

- Have a good understanding and knowledge of Storage Area Networks, Fibre Channel fundamentals and a basic understanding of TCP/IP and other networking protocols (DNS/NTP).
- Have experience creating Storage LUNs, presenting/exporting LUNs to a server and formatting the LUNs to make them usable for applications.
- Be able to troubleshoot hardware and software issues using logs and documentation.

Configuration Information

If the installer does not meet the profile or is not comfortable with the self-installation process, Hewlett Packard Enterprise recommends engaging the Hewlett Packard Enterprise sales representative or Hewlett Packard Enterprise Channel Partner to purchase HPE Deployment Services.

Customer responsibilities

- Ensure that the host and SAN environment is supported and compliant with HPE recommendations and best practices. Host and SAN Implementation Guides are available at <https://support.HPE.com/hpesc/public/home>. Support Matrix are available on SPOCK (HPE Storage Single Point of Connectivity Knowledge) <http://www.HPE.com/storage/spock>.
- Resolve any problems with their SAN and host environment prior to installing the HPE Alletra Storage MP B10000.

Notes: Customers performing a self-install (according to rules identified above) will not void their warranties and will be fully supported.

HPE Alletra Software MP B10000 Upgrades

When upgrading Controller Nodes or NVMe SSDs to an existing HPE Alletra Storage MP B10000, use the upgrade products shown below.

HPE Alletra Storage MP B10000 Controller Node Field Upgrade	SKU
HPE Alletra Storage MP B10130 Controller Node Field Upgrade	SOR21B
HPE Alletra Storage MP B10130 TAA-compliant Controller Node Field Upgrade	SOS41B
HPE Alletra Storage MP B10140 Controller Node Field Upgrade	S4V66B
HPE Alletra Storage MP B10140 TAA-compliant Controller Node Field Upgrade	S4V67B
HPE Alletra Storage MP B10230 Controller Node Field Upgrade	R7D02B
HPE Alletra Storage MP B10230 TAA-compliant Controller Node Field Upgrade	SOS42B
HPE Alletra Storage MP B10240 Controller Node Field Upgrade	R7D03B
HPE Alletra Storage MP B10240 TAA-compliant Controller Node Field Upgrade	SOR20B

HPE Alletra Storage MP B10000 Upgrade SSDs	SKU
HPE Alletra Storage MP 1.92 TB NVMe Self-encrypting Upgrade SSD	R9H66C
HPE Alletra Storage MP 3.84 TB NVMe Self-encrypting Upgrade SSD	R9H67C
HPE Alletra Storage MP 7.68 TB NVMe Self-encrypting Upgrade SSD	R9H68C
HPE Alletra Storage MP 15.36 TB NVMe Self-encrypting Upgrade SSD	R9H69C
HPE Alletra Storage MP 30.72 TB NVMe Self-encrypting Upgrade SSD	S1Z68C
HPE Alletra Storage MP 1.92 TB NVMe FIPS 140-3 Encrypted Upgrade SSD	R9H70C
HPE Alletra Storage MP 3.84 TB NVMe FIPS 140-3 Encrypted Upgrade SSD	R9H71C
HPE Alletra Storage MP 7.68 TB NVMe FIPS 140-3 Encrypted Upgrade SSD	R9H72C
HPE Alletra Storage MP 15.36 TB NVMe FIPS 140-3 Encrypted Upgrade SSD	R9H73C
HPE Alletra Storage MP 30.72 TB NVMe FIPS 140-3 Encrypted Upgrade SSD	S4P49C
HPE Alletra Storage MP 1.92 TB NVMe FIPS 140-3 Encrypted TAA-compliant Upgrade SSD	S0Q01C
HPE Alletra Storage MP 3.84 TB NVMe FIPS 140-3 Encrypted TAA-compliant Upgrade SSD	S0Q02C
HPE Alletra Storage MP 7.68 TB NVMe FIPS 140-3 Encrypted TAA-compliant Upgrade SSD	S0Q03C

Configuration Information

Description	SKU
HPE Alletra Storage MP 15.36 TB NVMe FIPS 140-3 Encrypted TAA-compliant Upgrade SSD	S0Q04C
HPE Alletra Storage MP 15.36 TB NVMe QLC Upgrade SSD	S3L73C
HPE Alletra Storage MP 30.72 TB NVMe QLC Upgrade SSD	S3L74C

Technical Specifications

HPE Alletra Storage MP B10000 Specifications

Physical Dimensions	Width in/mm	Depth in/mm	Height in/mm/U	Weight lb/kg
HPE 42U 1075 mm G2 Advanced Series Rack	23.50/ 597	43.78/ 1111	78.99/ 2006	281/ 127
HPE 42U 1075 mm G2 Enterprise Series Rack	23.54/ 598	44.30/ 1125	78.98/ 2007	230/ 105
HPE 42U 1200 mm G2 Advanced Series Rack	23.50/ 597	50.65/ 1286	78.99/ 2006	311/ 141
HPE 42U 1200 mm G2 Enterprise Series Rack	23.54/ 598	51.19/ 1300	78.98/ 2007	251/ 114
HPE Alletra Storage MP B10000 Base Enclosure (Enclosure, two Controller IOMs, two PS, one CDM, no drives, no HBAs)	19.00/ 483	33.11/ 841	3.44/ 87.5/ 2	74.0/ 33.6
HPE Alletra Storage MP B10000 Base Enclosure Packaging Pallet	23.00/584 24.00/610	38.75/984 40.00/1016	11.50/292	
HPE Alletra Storage MP NVMe SSD with carrier	3.11/ 79	6.57/ 167	0.59/ 15.0	0.63/ 0.28
HPE 32 GB 4-port FC Host Bus Adapter	3.33/ 84.6	5.20/ 132	0.59/ 15.0	0.50/ 0.23
HPE 10/25 GB 4-port Ethernet Host Bus Adapter	3.33/ 84.6	5.20/ 132	0.59/ 15.0	0.50/ 0.23

Power Requirements

Input Voltage - AC PCM options

- HPE Alletra Storage MP B10000:
 - High-Line: 200 to 240 VAC (50 to 60 Hz)
 - Low-Line: 100 to 120 VAC (50 to 60 Hz)
 - DC: -48 VDC

Notes:

- Refer to the [Power Supplies](#) section to find the proper power supplies for the intended infrastructure
- Refer to the [HPE Power Advisor](#) online tool or [Ninja Online](#) for power consumption, heat loading, and circuit sizing information.

Technical Specifications

Environmental Specifications

Operating Temperature	41°F to 95°F (5°C to 35°C) - Reduce the rating by 1°F for each 1,000 ft altitude (1.8°C/1,000 m)	
	– Standard – with 30 TB QLC SSDs	
Shipping Temperature	-30°C to 60°C (-22°F to 140°F). The maximum temperature change rate is 20°C per hour (36°F per hour).	
Operating Altitude (ft/m) max.	10,000 ft/ 3,048 m	
Shipping Altitude (ft/m) max.	40,000ft/ 12,192 m	
Humidity	10% to 90% non-condensing	
Shipping Humidity	10% to 90% non-condensing	
Operating Vibration	0.25 G, Sine, 5-500 Hz; 0.25 GRMS, Random 5-500 Hz	
Non-operating Vibration	0.75 G, Sine, 5-500 Hz	
Operating Shock	5G, 11ms, half-sine	
Non-operating Shock	10 G, 11ms, half-sine	
Maximum Exhaust Air Flow	HPE Alletra Storage MP Base: 267 CFM	
Acoustic Sound Pressure Level*	Typical, 27C Ambient**	Maximum, 100% Duty Cycle Fans
HPE Alletra Storage MP Base	58.9 dBA to 72 dBA	78.3 dBA
Notes: **Typical Acoustic emission varies based on hardware configuration and options		

Electromagnetic Compatibility

- CISPR 32:2015/ EN 55032: 2015 +A11:2020 Class A
- BS EN 55032:2015 +A11:2020
- CISPR 35:2016/ EN 55035:2017 +A11:2020
- BS EN 55035:2017 +A11:2020
- IEN 61000-3-2: 2019 +A1:2021
- EN 61000-3-3: 2013 +A2:2021
- AS/NZS CISPR 32:2015 +A1:2020 Class A
- CNS 13438:2006 Class A
- 47 CFR Part 15 Subpart b Class A
- ICES-003 Issue 7 Class A
- VCCI-CISPR 32: 2016 Class A
- RRA Notice No. 2021-3 (2021.02.08) Class A
- RRA Notice No. 2021-10 (2021.02.08)

Safety

- IEC 60950-1:2005 (2nd Edition); +A1:2009 +A2:2013
- EN 62479:2010
- IEC 62368-1: 2014/ IEC 62368-1:2018
- EN 62368-1:2014+A11:2017/ EN 62368-1:2020 +A11:2020
- CNS 14336-1
- ANSI/UL 62368-1:2021
- CAN/CSA-C22.2 No. 62368-1:19 Update No. 1-2021

Technical Specifications

Certifications/Markings

- BIS
 - BSMI
 - cCSAus
 - CE
 - ENERGY STAR (B10140)
 - FCC Class A
 - IC Class A
 - KCC
 - Morocco
 - NEBS Level 3
 - RCM
 - VCCI
 - WEEE
 - China RoHS
 - EU RoHS
 - UKCA
-

Summary of Changes

Date	Version History	Action	Description of Change
02-Mar-2026	Version 29	Changed	Updated Step 9: Choose software to include Renewal section. - Added new optional SSD offerings in Step 4: Choose drives, including FIPS-encrypted and self-encrypting NVMe SFF SSDs (7.68 TB and 15.36 TB). - Added HPE Alletra Storage MP B10000 per TB Renewal Software and Support SaaS (S6J58AAE) to Step 9: Choose software.
19-Jan-2026	Version 28	Changed	Visual rebranding only—updated typography, colors, and design elements to align with new HPE brand standards. No technical specifications or content were modified.
01-Dec-2025	Version 27	Changed	Removed reference to “See HPE Store More Guarantee for more information” from the <i>Run any app – without compromise with a 100% data availability guarantee</i> section.
03-Nov-2025	Version 26	Changed	Overview, Standard Features and Configuration Information sections were updated - The new HPE Alletra Storage MP B10140 (4N) is available in a switchless. New purchase of B10000 Array information included.
08-Sep-2025	Version 25	Changed	Configuration Information section was updated - Cables per node SKUs were updated in Requirements for 100 GbE backend connectivity section.
		Removed	Obsolete SKUs were removed - SOS39A, SOS43A.
25-Aug-2025	Version 24	Changed	Configuration information section was updated - Notes on Power supplies section were updated.
		Added	Power Cord SKUs were added. R9S02A, R9R66A, R9S03A, R9R67A.
04-Aug-2025	Version 23	Changed	Configuration information section was updated - Notes on Racking Options Step 8 were updated.
		Added	SOR21B, SOS41B, S4V66B, S4V67B, R7D02B, SOS42B, R7D03B, SOR20B, R9H66C, R9H67C, R9H68C, R9H69C, S1Z68C, R9H70C, R9H71C, R9H72C, R9H73C, S4P49C, SOQ01C, SOQ02C, SOQ03C, SOQ04C, S3L73C, S3L74C.
07-Jul-2025	Version 22	Changed	Overview section was updated – New Features added Standard Features section was updated- Switches Information table updated Technical Specifications section was updated - Certifications/Markings.
02-Jun-2025	Version 21	Changed	Configuration Information section was updated Added new 30 TB FIPS SSD information.
28-Apr-2025	Version 20	Changed	Configuration Information and Technical Specifications sections were updated.
07-Apr-2025	Version 19	Changed	Overview, Service and Support and Configuration Information sections were updated.
03-Feb-2025	Version 18	Changed	Configuration Information section was updated.
16-Dec-2024	Version 17	Changed	Configuration Information section was updated.

Summary of Changes

Date	Version History	Action	Description of Change
02-Dec-2024	Version 16	Changed	Service and Support and Configuration Information sections were updated.
20-Nov-2024	Version 15	Changed	Overview, Standard Features, Service and Support and Configuration Information sections were updated. HPE Product Name Rebranding.
21-Oct-2024	Version 14	Changed	Overview, Standard Features, and Configuration Information sections were updated.
07-Oct-2024	Version 13	Changed	Standard Features, Configuration Information and Technical Specifications sections were updated.
19-Aug-2024	Version 12	Changed	Standard Features and Configuration Information sections were updated.
05-Aug-2024	Version 11	Changed	Overview, Standard Features, and Configuration Information sections were updated.
03-Jun-2024	Version 10	Changed	Configuration Information section was updated
26-Feb-2024	Version 9	Changed	Networking product names were updated.
05-Feb-2024	Version 8	Changed	Overview, Standard Features, and Configuration Information sections were updated.
08-Jan-2024	Version 7	Changed	Configuration Information section was updated
13-Nov-2023	Version 6	Changed	Overview, Standard Features, Service and Support and Configuration Information sections were updated. HPE Services Rebranding
02-Oct-2023	Version 5	Changed	Standard Features and Configuration Information sections were updated
07-Aug-2023	Version 4	Changed	Configuration Information and Technical Specifications sections were updated
10-Jul-2023	Version 3	Changed	Configuration Information and Technical Specifications sections were updated
15-May-2023	Version 2	Changed	Configuration Information section was updated
03-Apr-2023	Version 1	New	New QuickSpecs.

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